Play any song by ear

A Learn page that teaches a seven-step method for working out a song by listening. It has playable examples for every step, two traditional songs worked out stage by stage, four scored ear drills and a cheat sheet.

Builds on the Theory map

This page reuses the Theory map's player, fretboard, grid and block visuals, example buttons and lane tokens. Build the Theory map first (its own prompt and spec), then this. Use the app's existing components and fonts where they exist.

Files that come with it

- `cursor-prompt.md` : the build instructions
- `ear.ts` : all content, worked songs, drills (seeded, testable), tables, tap-tempo helpers
- `ear.test.ts` : 10 tests for that file
- `reference-prototype.html` : open in a browser to see and hear the behaviour

Contents

1. Page structure
2. Hero and step strip
3. Method step with a fretboard example
4. Chord step with the key chart
5. Rhythm step with tap tempo
6. Worked song
7. Ear drills
8. Cheat sheet
9. Phone layout
10. Dark theme
11. Acceptance checklist

Prototype screenshots, 25 October 2026

1 · STRUCTURE

One page, five sections

Route suggestion: `/learn/by-ear`, linked from the Learn page, the main menu and the Theory map. Client-side only.

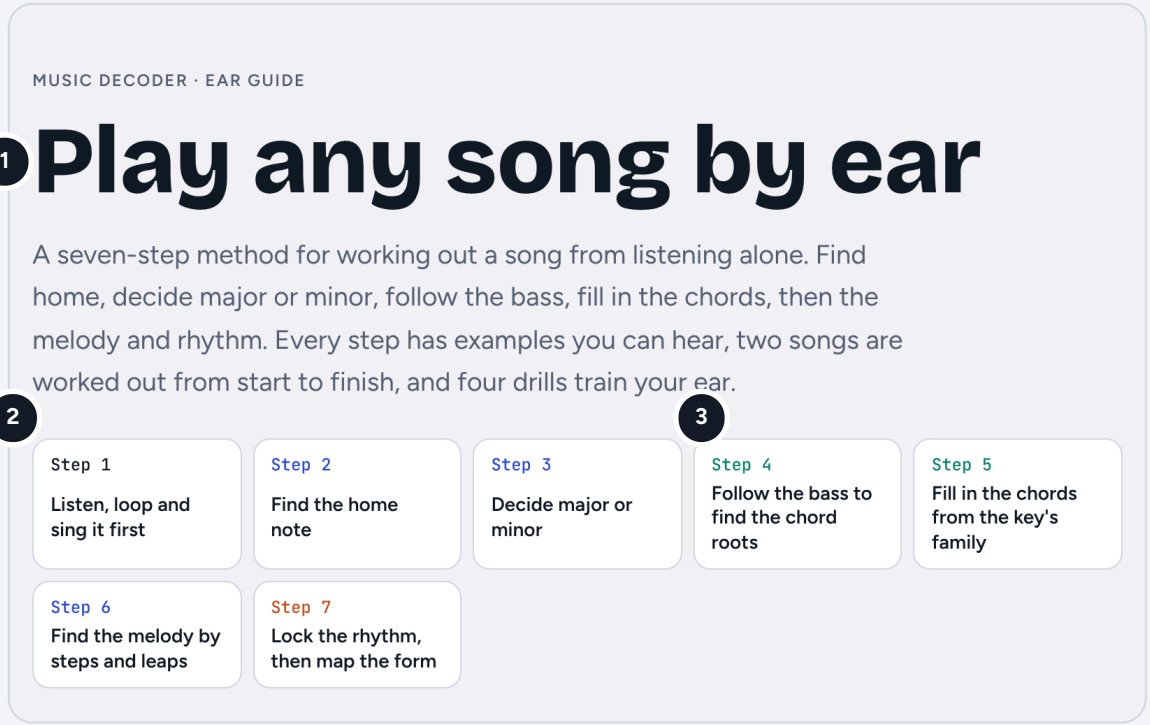
Hero	**The method**	**Worked examples**	**Ear drills**	**Cheat sheet**
title, lead, 7-step strip	7 step rows, examples + visual	2 songs, stages + sticky visual	4 scored, random drills	progressions + fixes

PIECE	REUSE FROM THE THEORY MAP	NEW FOR THIS PAGE
Audio	Shared player (`getTheoryPlayer()`), one sound at a time app-wide	Widen `play()` to accept `Sound` (bpm, lengthBeats, events) for drills, which have no visual
Visuals	FretVisual, GridVisual, BlocksVisual and the `data-hl` highlight approach	None
Buttons	ExampleButton (play ↔ stop, `aria-pressed`)	Drill answer buttons with right/wrong states
Layout	Two-column row card (text left, examples + visual right), stacks under 820px	Step strip, worked-song card with a sticky visual, 2-column drill grid, key chart, tap-tempo pad
Tokens	Lane colours and tints, amber highlight	Drill states: right = chords lane (teal), wrong = time lane (orange)

2 · HERO

Title, one-paragraph promise, and the seven steps

The strip is a real sequence, so it is numbered. Each card jumps to its step row.

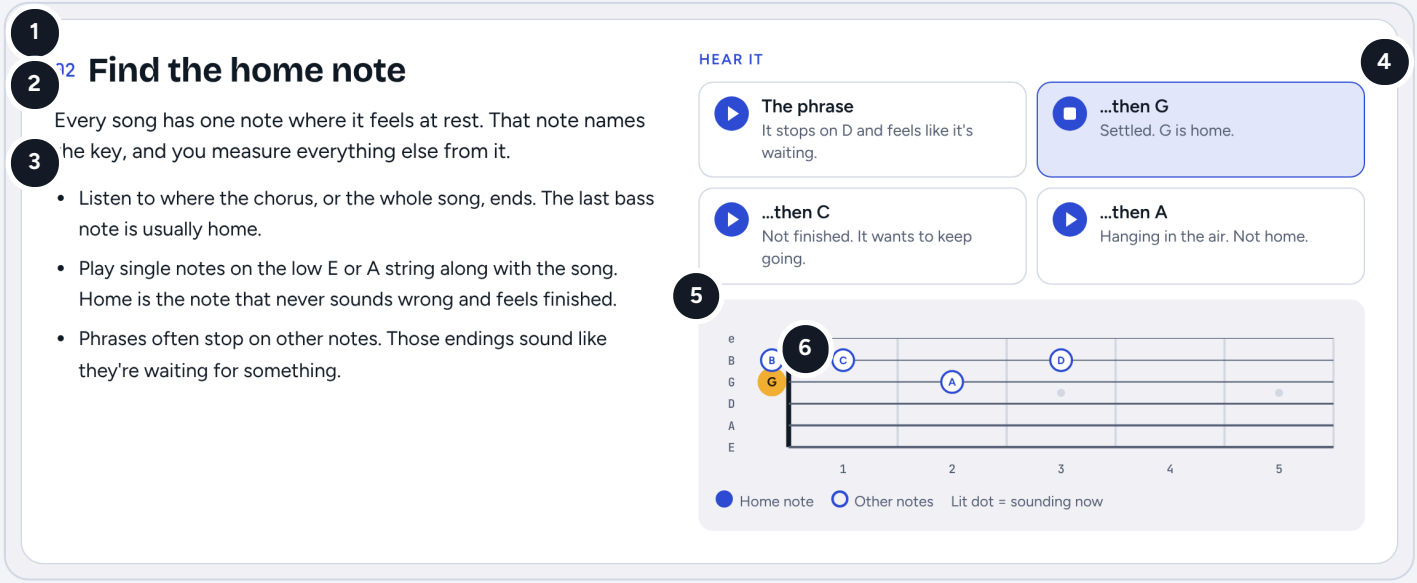


1. **Title and lead**
   Display font. The lead names the method and what the page contains.
2. **Step card**
   "Step n" in mono, coloured by the step's lane, then the step title. Links to `#step-<id>`.
3. **Lane colour**
   Pitch steps blue, chord steps teal, time orange, the song/listen step ink. Cards are at least 130px wide and wrap to fill the hero width (two per row on phone).

3 · METHOD STEP

Text on the left, hear it on the right

Every step uses this row: `EAR_STEPS[i]` gives the number, title, why, how and examples. Shown: step 2 with "...then G" playing.



1. **Number and title**
   Two-digit number in mono and the lane colour, then the title.
2. **Why**
   One or two sentences, slightly larger than body text.
3. **How**
   Three practical bullets for doing it with a real song.
4. **Example buttons**
   Same ExampleButton as the Theory map. Pressing shows that example's visual and plays it; pressing again stops.
5. **Visual**
   The pressed example's visual (fret, grid or blocks).
6. **Live note**
   The sounding note lights amber. Here the final G, the home note.

4 · CHORD STEP

Key chart plus a chord-fit example

`extra: "keyChart"` adds the chart under the bullets. Data: `KEY_CHART`.

05 Fill in the chords from the key's family

Songs mostly use the chords that belong to their key. Once you know the key, there are only about six chords to try.

- Find the key's row in the chart below. Try I, IV, V and vi first (highlighted).
- If the bass plays A in C major, try Am (vi) before A major.
- Check against the melody: the note on beat 1 is usually one of the chord's notes.

MAJOR KEY	I	ii	III	IV	V	VI
G	G	Am	Bm	C	D	Em
C	C	Dm	Em	F	G	Am
D	D	Em	F♯m	G	A	Bm
A	A	Bm	C♯m	D	E	F♯m
E	E	F♯m	G♯m	A	B	C♯m
Minor key	i	III	iv	v	♭VI	♭VII
Am	Am	C	Dm	Em	F	G
Em	Em	G	Am	Bm	C	D

Highlighted chords are the ones to try first. Each minor key shares its chords with a major key: Am with C, Em with G.

HEAR IT

One melody note, three chords
F sits inside F and Dm. Over C it rubs against the chord's E.

The G family: I – IV – V – vi
G, C, D and Em are the first four chords to try in G.

Melody	F	F	F
Chords	C I	F IV	Dm ii

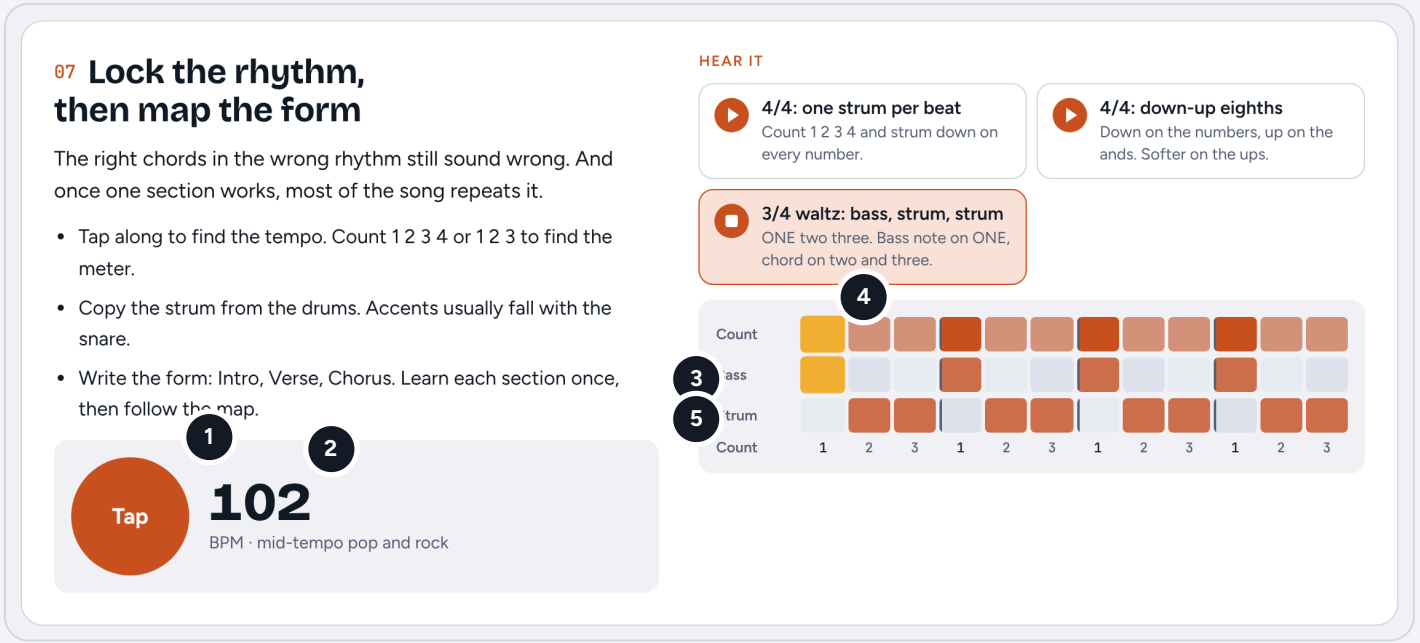
1. **Try-first cells**
   I, IV, V and vi (major) and i, iv, ♭VI, ♭VII (minor) are tinted teal. Indices in `tryFirstMajor` and `tryFirstMinor`.
2. **Minor rows**
   Separated by a thicker rule with their own numeral header (i III iv v ♭VI ♭VII).
3. **Melody row**
   The "one note, three chords" example holds the same F over C, F and Dm.
4. **Lit block**
   The chord that is sounding.

The chart scrolls sideways inside its well on phones (min width 520px).

5 · RHYTHM STEP

Tap tempo and strum patterns

`extra: "tapTempo"` adds the tap pad. Strum rows sound a full chord on each hit and light their cell.

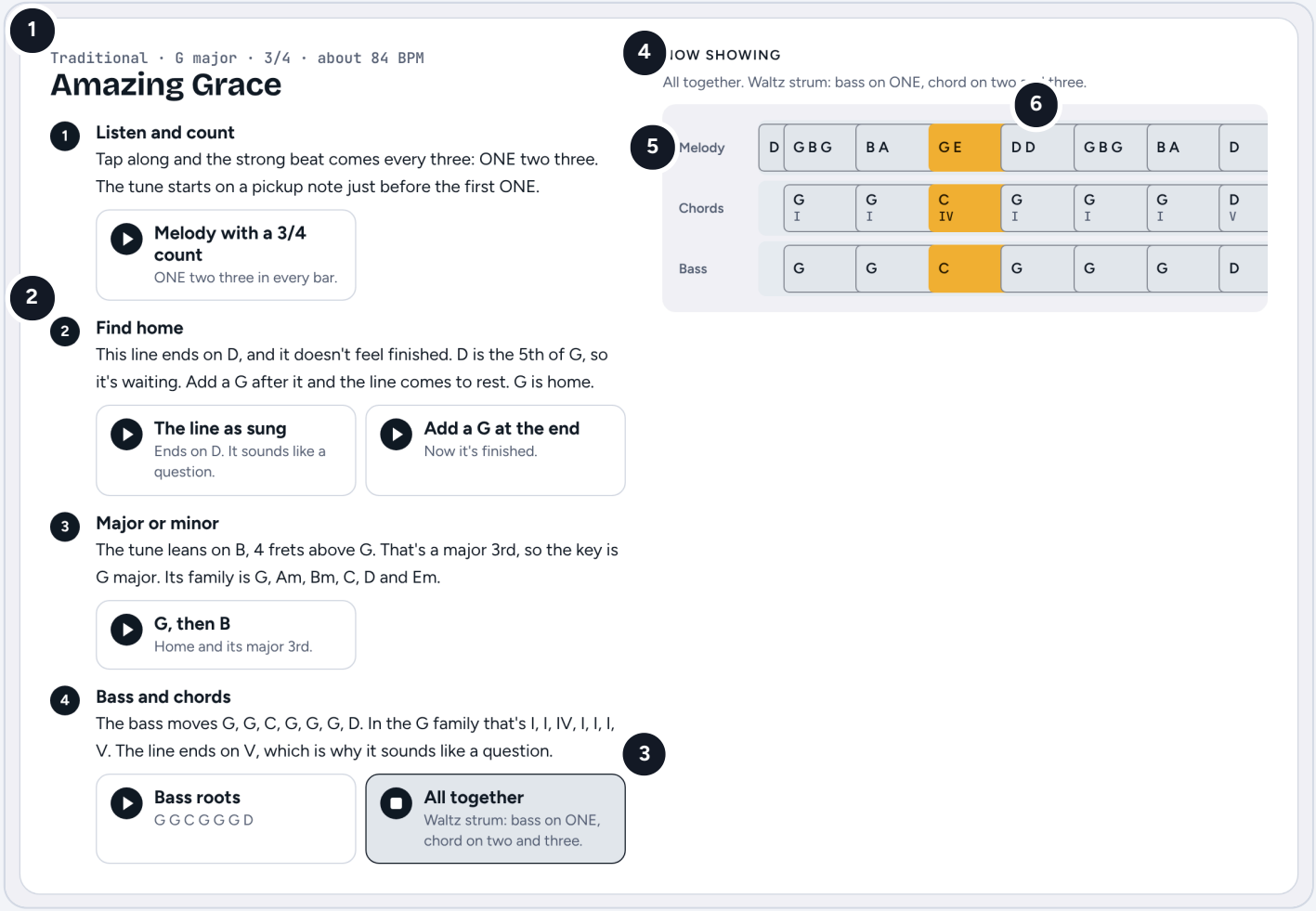


1. **Tap pad**
   Large round button in the time colour. Works with click, touch and Space/Enter.
2. **Readout**
   BPM from `tempoFromTaps()` (last 8 taps; reset after 2 s idle) and `tempoLabel()`. Shows … after the first tap.
3. **Strum row**
   Grid row whose hits strum a chord (built by `strumGridExample`).
4. **Lit cell**
   Amber while sounding.
5. **Count row**
   1 2 3 for 3/4, 1 + 2 + for 8ths.

6 · WORKED SONG

Stages on the left, one shared visual on the right

Data: `WORKED_SONGS`. Each stage has its own example buttons; all of them drive one visual that stays in view.



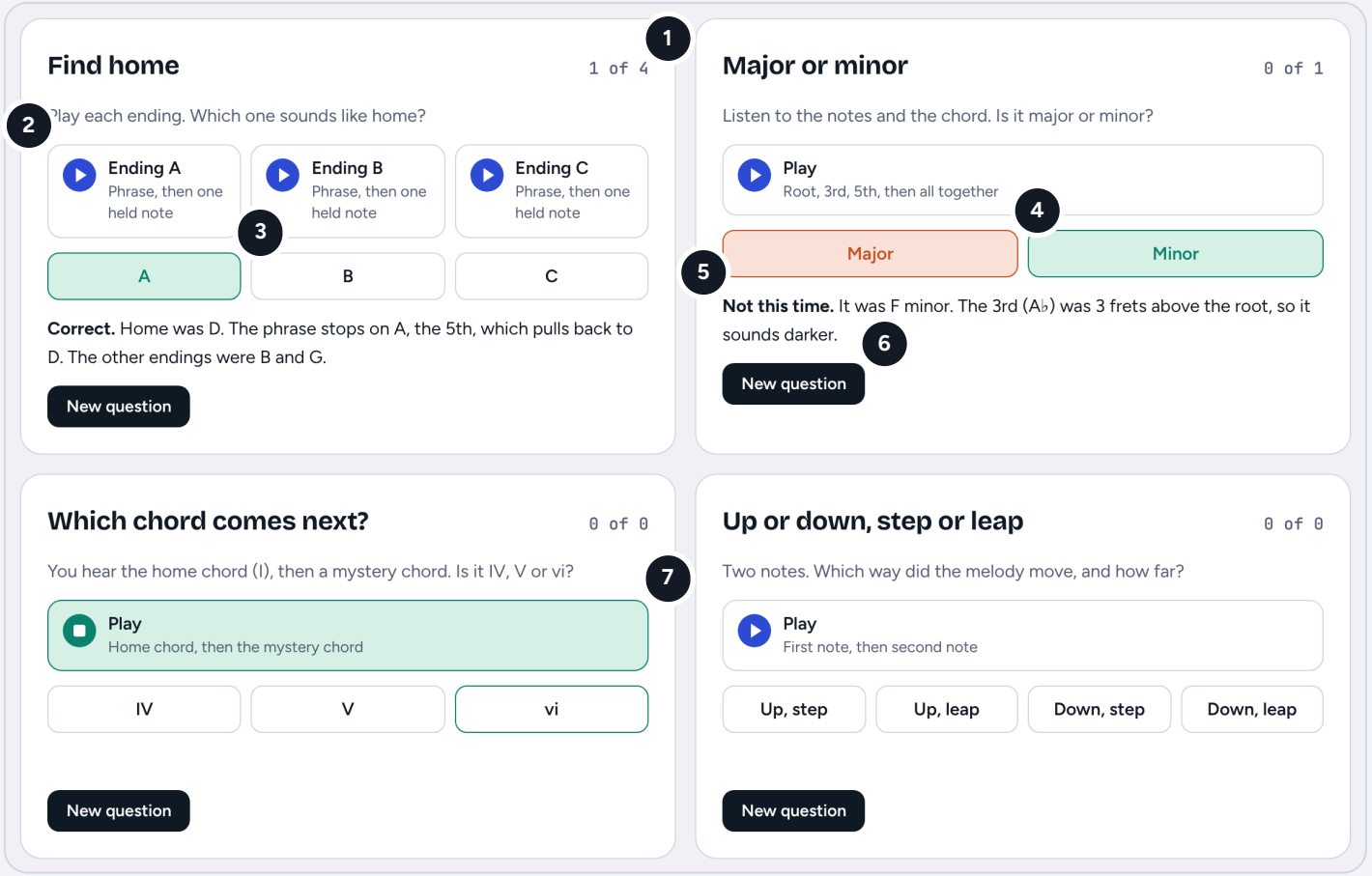
1. **Meta line**
   Traditional · key · meter · tempo, in mono.
2. **Stage**
   Numbered circle, stage title, text, then its example buttons.
3. **Playing**
   Same play/stop behaviour as everywhere else.
4. **Caption**
   "<label>. <sub>" of the example being shown.
5. **Layer rows**
   Melody (note names), Chords (name + numeral), Bass. The 1-beat pickup bar is a narrow first block.
6. **Lit blocks**
   Every layer in the current bar lights at once.

The visual column is sticky on desktop (top: safe-area inset + 12px) and becomes static when stacked.

7 · EAR DRILLS

Four random drills with a score

Data: `DRILLS`. `generate(rng)` returns listen sounds, options (exactly one correct) and an explanation.



1. **Score**
   "right of total" for this visit. Optional: keep it in the app store with practice history.
2. **Listen buttons**
   Audition as often as you like. Find home has three endings (A, B, C); the others have one Play button.
3. **Right answer**
   After answering, all options lock and the correct one turns teal.
4. **Wrong pick**
   The chosen wrong option turns orange.
5. **Feedback**
   "Correct." or "Not this time." in bold, then the explanation. `aria-live="polite"`.
6. **New question**
   Stops any sound and generates a fresh question.
7. **Playing**
   Listen buttons share the one-at-a-time player.

8 · CHEAT SHEET

Progressions and fixes

Data: `COMMON_PROGRESSIONS` and `TROUBLESHOOTING`. Two columns on desktop, stacked on phone.

Progressions you'll hear again and again

When you have the first two chords, check this list. The rest of the loop is often one of these.

NAME	NUMBERS	IN G / EM	IN C / AM	WHERE YOU HEAR IT
Three-chord	I – IV – V	G C D	C F G	Folk, country, blues, early rock
Pop loop	I – V – vi – IV	G D Em C	C G Am F	Pop, ballads, singalongs
Pop loop, sad start	vi – IV – I – V	Em C G D	Am F C G	Emotional pop and rock
Fifties	I – vi – IV – V	G Em C D	C Am F G	Doo-wop and oldies
12-bar blues	I I I I · IV IV I I · V IV I V	G7 C7 D7	C7 F7 G7	Blues, rock'n'roll
Minor rock	i – ♭VII – ♭VI – ♭VII	Em D C D	Am G F G	Rock and metal
Descending minor	i – ♭VII – ♭VI – V	Em D C B	Am G F E	Flamenco, surf, dramatic rock

When it doesn't sound right

The chord is close but not right

Try its relative: C ↔ Am, G ↔ Em, F ↔ Dm. They share two notes.

The bass note fits but the chord sounds off

Switch between major and minor, or add a 7th (G → G7).

Nothing fits anywhere

The recording may be tuned down a half step, which is common in rock. Tune down, or move everything up one fret.

It worked until the chorus

Some songs change key for a section. Find the new home note for that part.

The melody note clashes on beat 1

The chord is probably different there. Look for a chord that contains that note.

You're lost

Go back to the home note, loop four bars at 70%, and rebuild from the bass.

1. **Numbers**
   Roman numerals in mono so the shape reads at a glance.
2. **In G / In C**
   Same progression in a sharp-friendly and a flat-friendly key (minor ones in Em and Am).
3. **Fix**
   Problem in bold, fix underneath. Six items.

9 · PHONE (390 PX)

Everything stacks

No horizontal page scroll. Only the key chart, wide block visuals and the progressions table scroll inside their own containers.

MUSIC DECODER · EAR GUIDE

Play any song by ear

A seven-step method for working out a song from listening alone. Find home, decide major or minor, follow the bass, fill in the chords, then the melody and rhythm. Every step has examples you can hear, two songs are worked out from start to finish, and four drills train your ear.

1

Step 1
Listen, loop and sing it first

Step 2
Find the home note

Step 3
Decide major or minor

Step 4
Follow the bass to find the chord roots

Step 5
Fill in the chords from the key's family

Step 6
Find the melody by steps and leaps

Step 7
Lock the rhythm, then map the form

Hero and strip

02 Find the home note

Every song has one note where it feels at rest. That note names the key, and you measure everything else from it.

- Listen to where the chorus, or the whole song, ends. The last bass note is usually home.
- Play single notes on the low E or A string along with the song. Home is the note that never sounds wrong and feels finished.
- Phrases often stop on other notes. Those endings sound like they're waiting for something.

HEAR IT
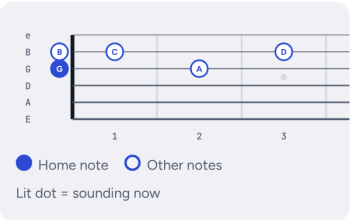
1

The phrase
It stops on D and feels like it's waiting.

...then G
Settled. G is home.

...then C
Not finished. It wants to keep going.

...then A
Hanging in the air. Not home.



Step row, stacked

Find home

0 of 0

Play each ending. Which one sounds like home?

Ending A
Phrase, then one held note

Ending B
Phrase, then one held note

Ending C
Phrase, then one held note

1

A

B

C

New question

Drill card

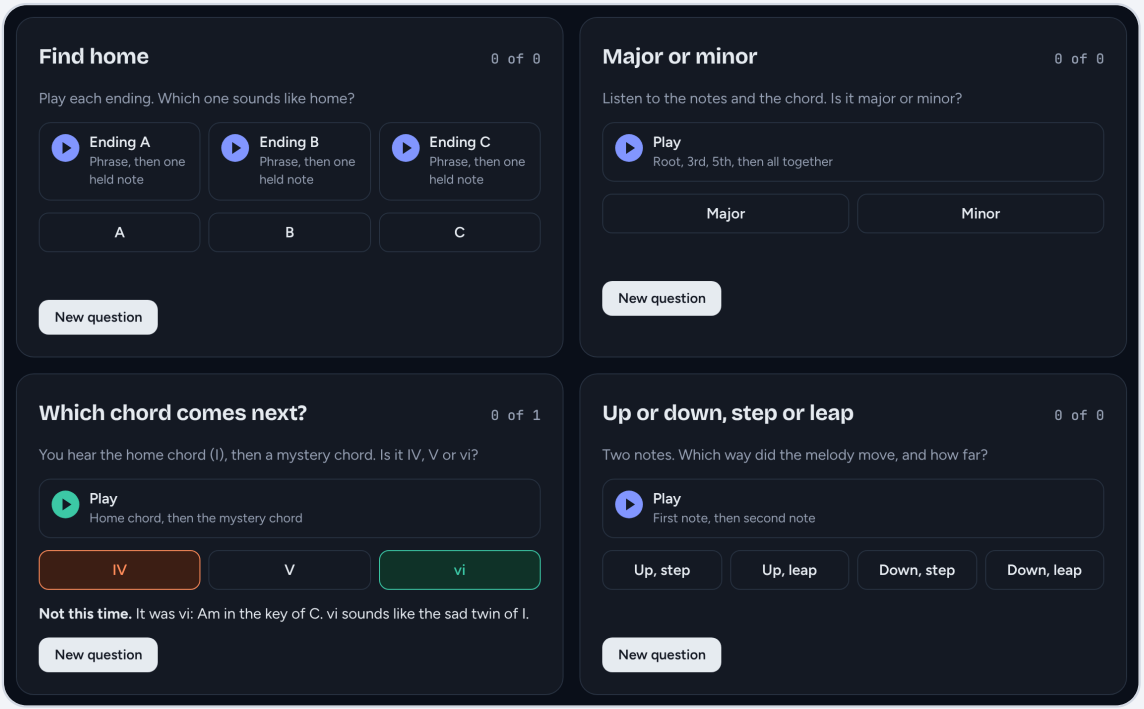
1 **Layout**
Strip in two columns; rows, worked songs and drills in one column; example buttons fill the width.

10 · DARK THEME

Same layout and lane colours as the Theory map in dark

Right and wrong answer states use the lane tints, so they adapt automatically.





1 Tokens

No new colours. Reuse the Theory map's dark lane values.

11 · ACCEPTANCE

Done when

The same checklist is in `cursor-prompt.md`.

1. Route renders without errors; no horizontal page scroll at 390 px.

2. All 31 examples (19 in the method, 12 in the worked songs) play through the app's audio engine, light their visual in sync, and stop when pressed again or when anything else starts.

3. The step strip jumps to each step; the 70% example is audibly slower at the same pitch.

4. Key chart highlights the try-first columns; tap tempo shows BPM after two taps and resets after 2 s idle.

5. Worked songs: every stage button updates the shared visual and caption; the visual stays in view on desktop.

6. Drills: listen, answer, lock, colour, explain, score and new question all work; exactly one correct option each time; New question stops sound.

7. Keyboard: every button reachable with visible focus; tap pad works with Space/Enter.

8. Reduced motion: no scaling; highlights still change colour. Leaving the page stops audio.

9. Unit tests: ear.test.ts passes, plus a component test for the drill flow. Browser tests: play an example, answer a drill, tap tempo, phone layout.